SAT Math Mastery · Session 11

Area and Volume

Time	3 hours: Part A in class (about 2 hours of solving), then Part B, the self-assessment, done independently (about 50 minutes), and a short error log.
Topics	area of composite figures · volume of prisms, cylinders, cones, spheres and pyramids · surface area · composite and carved solids · similar figures and solids (scale factors for length, area, volume) · right triangles inside solids · unit conversion · displacement and packing
Name / Date	_____ _____

PART A · IN CLASS — 35 PROBLEMS

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

1. HARD · DESMOS

A scale model of a section of a long building is geometrically similar to the actual structure, with a scale factor of 1 to 20 (model to actual). If the volume of the model is 3 cubic meters, what is the volume of the actual segment, in cubic meters?

(A) 8,000

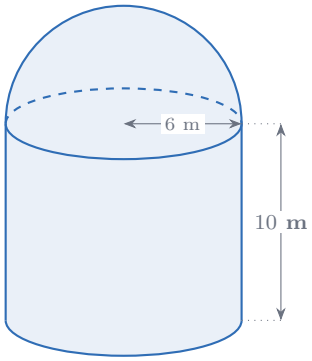
(B) 24,000

(C) 60

(D) 1,200

2. Hard · Desmos

A pavilion is designed as a cylindrical base topped by a hemispherical dome of the same radius. The cylindrical base has radius 6 meters and height 10 meters, and the dome is a hemisphere with radius 6 meters. What is the total volume of the structure, in cubic meters, in terms of π?



(A) 360π

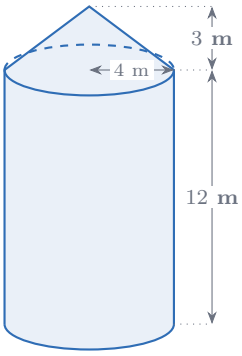
(B) 432π

(C) 504π

(D) 576π

3. Hard · Desmos

A grain silo is designed as a cylindrical base topped by a conical roof of the same radius. The cylindrical base has radius 4 meters and height 12 meters, and the conical roof has radius 4 meters and height 3 meters. What is the total volume of the structure, in cubic meters, in terms of π?



(A) 208π

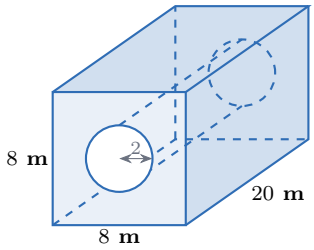
(B) 240π

(C) 176π

(D) 192π

4. Hard · Desmos

A stone block used in a restoration project is a rectangular prism measuring 8 meters by 8 meters by 20 meters, with a cylindrical channel of radius 2 meters drilled completely through its 20-meter length. What is the volume of the remaining stone, in cubic meters, in terms of π?



(A) $1280 - 160\pi$

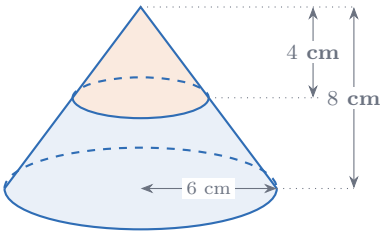
(B) $1280 - 20\pi$

(C) $1200 - 80\pi$

(D) $1280 - 80\pi$

5. Hard · Desmos

A cone has base radius 6 centimeters and height 8 centimeters, apex pointing up. A plane parallel to the base cuts the cone at a height of 4 centimeters from the apex, removing the smaller cone on top. What is the volume of the removed smaller cone, in cubic centimeters, in terms of π?



(A) 48π

(B) 96π

(C) 12π

(D) 24π

6. HARD · DESMOS

A cylindrical water tank has a radius of 5 meters and a height of 12 meters. What is the total surface area of the tank, including both circular bases, in square meters, in terms of π?

(A) 150π

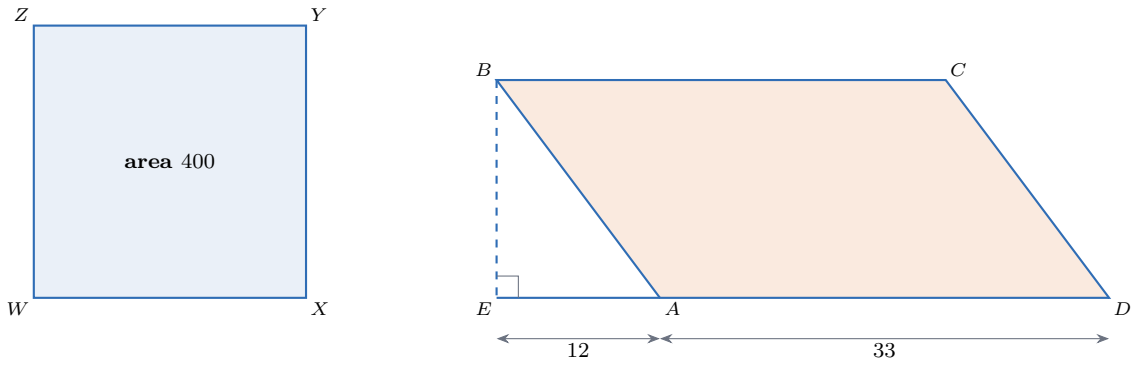
(B) 170π

(C) 220π

(D) 120π

7. HARD · DESMOS

In the figure, square $WXYZ$ has an area of 400 square units. Side AB of parallelogram $ABCD$ has the same length as a side of the square. Right triangle ABE, formed by dropping an altitude from vertex B to the extension of side AD, has hypotenuse AB and one leg $AE = 12$. If side AD of the parallelogram has length 33, what is the area of parallelogram $ABCD$?



8. HARD · DESMOS

A rectangular vegetable garden originally has length s meters and width $0.4s$ meters. A surrounding path is widened, which decreases the garden's length by $0.1s$ and its width by $0.1s$. What is the percent decrease in the garden's area?

9. HARD · DESMOS

A hemisphere is half of a sphere. If a hemisphere has a radius of 18 inches, which of the following is closest to the volume, in cubic inches, of the hemisphere?

(A) 24,400

(B) 2,400

(C) 6,100

(D) 12,200

10. HARD · DESMOS

A conical grain storage silo has a radius of 6 meters and a height of 10 meters. Which of the following is closest to the volume of the silo, in cubic meters? (Use $\pi \approx 3.14$.)

(A) 376.8

(B) 452.2

(C) 565.2

(D) 282.6

11. HARD · DESMOS

A spherical water tank has a radius of 5 meters. Which of the following is closest to the surface area of the tank, in square meters? (Use $\pi \approx 3.14$ and the formula $SA = 4\pi r^2$.)

(A) 261.7

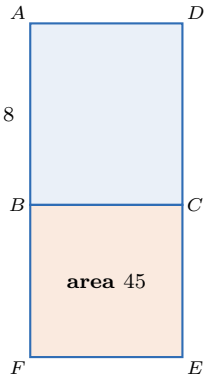
(B) 314.0

(C) 392.5

(D) 235.5

12. HARD · DESMOS

Side AB of rectangle $ABCD$ measures 8 units. Side BC of the rectangle is shared with side BC of square $BCEF$. If the area of square $BCEF$ is 45 square units, what is the area of rectangle $ABCD$?



(A) 80 square units

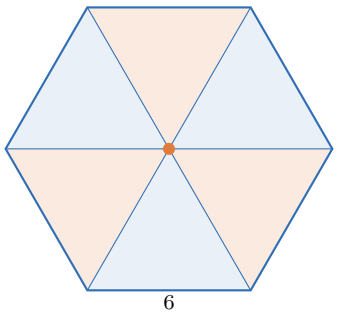
(B) $12\sqrt{5}$ square units

(C) $24\sqrt{5}$ square units

(D) $45 + 8\sqrt{5}$ square units

13. HARD · DESMOS

A hexagon is formed by arranging 6 equilateral triangles, each of which has a vertex at the center of the hexagon. If the length of one side of a triangle is 6, the area of the hexagon is $k\sqrt{3}$ square units. What is the value of k?



14. Hard · Desmos

Two similar rectangles have areas 630 and 2,520. The perimeter of the smaller rectangle is 210, and the perimeter of the larger is n. What is the value of n?

(A) 420

(B) 2,100

(C) 1,680

(D) 840

15. Hard · Desmos

A right square pyramid has a base area of 100 and a total surface area of $100 + 100\sqrt{5}$. What is the height of the pyramid?

16. Hard · Desmos

Two similar rectangular crates have surface areas of 144 square feet and 400 square feet. The smaller crate has a volume of 108 cubic feet. What is the volume, in cubic feet, of the larger crate?

(A) 1500

(B) 180

(C) 300

(D) 500

17. Hard · Desmos

Two similar solid blocks have surface areas of 208 square centimeters and 325 square centimeters. The larger block has a volume of 375 cubic centimeters. What is the volume, in cubic centimeters, of the smaller block?

18. Hard · Desmos

A closed right circular cylinder has a volume of 48π cubic centimeters and a total surface area of 56π square centimeters. The height of the cylinder is greater than its diameter. What is the height, in centimeters, of the cylinder?

(A) 3

(B) 4

(C) 12

(D) 2

19. Hard · Desmos

Sphere A has a great circle with circumference 8π centimeters. The surface area of sphere B is 80π square centimeters greater than the surface area of sphere A. What is the ratio of the volume of sphere B to the volume of sphere A?

(A) $\frac{9}{4}$

(B) $\frac{27}{8}$

(C) $\frac{5}{4}$

(D) $\frac{3}{2}$

20. Hard · Desmos

A right pyramid has a square base. The diagonal of the base is $10\sqrt{2}$ centimeters and the volume of the pyramid is 400 cubic centimeters. What is the lateral surface area, in square centimeters, of the pyramid (the four triangular faces, not the base)?

(A) 240

(B) 130

(C) 360

(D) 260

21. Hard · Desmos

A metal pipe is a hollow cylinder of length 15 centimeters and outer radius 10 centimeters. The wall has a uniform thickness of t centimeters, where $0 < t < 10$. The volume of metal in the pipe is 765π cubic centimeters. What is the value of t?

22. Hard · Desmos

A right pyramid has a rectangular base. Its height is 12 centimeters, and the slant heights of its triangular faces are 13 centimeters for one pair of opposite faces and 20 centimeters for the other pair. What is the total surface area, in square centimeters, of the pyramid, including the base?

(A) 936

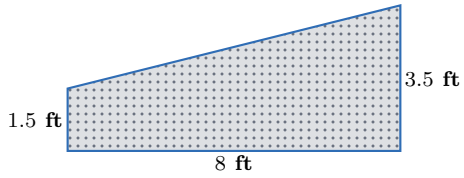
(B) 1160

(C) 616

(D) 866

23. Hard · Desmos

A raised garden bed is filled with soil. A vertical cross-section of the soil, perpendicular to the 12-foot length of the bed, is a trapezoid: the bed is 8 feet wide, and the soil is 1.5 feet deep along one side and 3.5 feet deep along the opposite side, with the surface sloping uniformly between them. Soil is sold at \$45 per cubic yard and billed for the exact volume (1 yard = 3 feet). What is the cost, in dollars, of the soil needed?



cross-section, perpendicular to the length of the bed

24. HARD · DESMOS

Cylinder P has radius 2 centimeters and height 6 centimeters. Cylinder Q is similar to cylinder P and has a total surface area of 200π square centimeters. What is the volume, in cubic centimeters, of cylinder Q?

(A) 150π

(B) 375π

(C) 600π

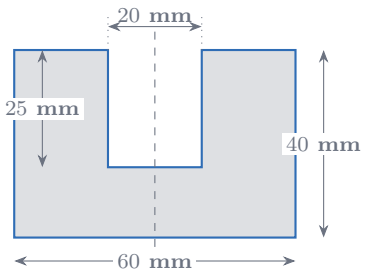
(D) 60π

25. HARD · DESMOS

Rectangular box P has dimensions 3 inches by 4 inches by 6 inches. Box Q is similar to box P and has a volume of 1125 cubic inches. What is the total surface area, in square inches, of box Q?

26. HARD · DESMOS

A brass bushing is a solid right circular cylinder with diameter 60 millimeters and height 40 millimeters. A cylindrical hole of diameter 20 millimeters is drilled into one flat face, along the axis, to a depth of 25 millimeters. The density of brass is 7.8 grams per cubic centimeter. Which of the following is closest to the mass of the bushing, in grams? (1 cm = 10 mm)



cross-section through the axis

(A) 82,090

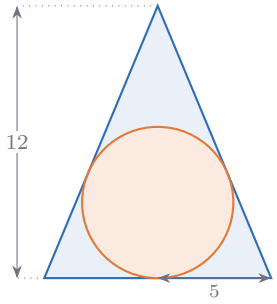
(B) 784

(C) 821

(D) 882

27. Hard · Desmos

A sphere is inscribed in a right circular cone whose base has radius 5 and whose height is 12, so that the sphere is tangent to the base and to the lateral surface of the cone. What fraction of the volume of the cone is occupied by the sphere? (Enter your answer as a fraction.)



28. Hard · Desmos

A museum has a small solid model of a sculpture and a full-size copy; the two are similar solids. The sealant needed to coat the whole surface is proportional to surface area. The model needs 1.6 liters of sealant and the full-size copy needs 14.4 liters. The model has a volume of 2.5 cubic feet. What is the volume, in cubic feet, of the full-size copy?

(A) 67.5

(B) 202.5

(C) 7.5

(D) 22.5

29. Hard · Desmos

Two aquariums are similar in shape. The smaller aquarium holds 250 liters and has 400 square decimeters of glass in total. The larger aquarium holds 432 liters. How many square decimeters of glass does the larger aquarium have in total? (Note that 1 liter = 1 cubic decimeter.)

30. Hard · Desmos

A roof ornament is a solid pyramid whose base is a square of area 100 square inches. The apex is directly above the center of the base, and each of the four edges from the apex to a corner of the base has length 13 inches. What is the combined area, in square inches, of the four triangular lateral faces?

(A) 260

(B) 120

(C) 340

(D) 240

31. Hard · Desmos

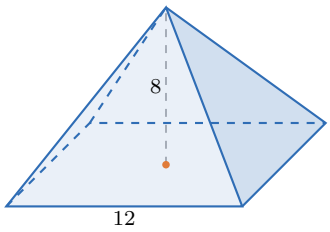
Two similar rectangular prisms have volumes in the ratio $8 : 27$. The smaller prism has a surface area of 40. What is the surface area of the larger prism?

32. Hard · Desmos

A rectangular tank has a base that is 40 centimeters by 30 centimeters. A solid cube with edge 12 centimeters is placed in the tank and sinks to the bottom, fully under water. By how many centimeters does the water level rise?

33. Hard · Desmos

A square pyramid has a base with side length 12 and a height of 8. What is the lateral surface area of the pyramid (the four triangular faces, excluding the base)?

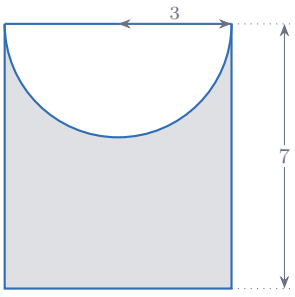


34. Hard · Desmos

The radius of a sphere is increased by 20%. By what percent does the volume of the sphere increase? (Enter as a decimal.)

35. Hard · Desmos

A solid is formed by carving a hemispherical bowl of radius 3 out of the top of a right circular cylinder of radius 3 and height 7. The volume of the solid is $k\pi$. What is the value of k?

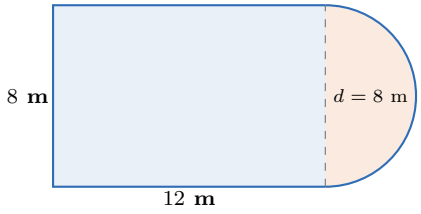


cross-section through the axis

PART B · SELF-ASSESSMENT — 22 PROBLEMS

36. Medium · Desmos

A school courtyard is shaped like a 12-meter by 8-meter rectangle with a semicircular reading nook attached to one of the 8-meter sides, as shown. The semicircle has a diameter of 8 meters. What is the total area of the courtyard, in square meters?



(A) 104π

(B) $96 + 4\pi$

(C) $96 + 8\pi$

(D) $96 + 16\pi$

37. Medium · Desmos

A cylindrical water tank supplying a public fountain has a radius of 15 meters and a volume of 4500π cubic meters. What is the height of the tank, in meters?

(A) 15

(B) 20

(C) 30

(D) 10

38. Medium · Desmos

A rectangular storage box has length 8 cm, width 5 cm, and height 4 cm. What is the total surface area of the box, in square centimeters?

(A) 92

(B) 144

(C) 160

(D) 184

39. Medium · Desmos

A cube has a total surface area of 150 square centimeters. What is the volume of the cube, in cubic centimeters?

(A) 125 (B) 150

(C) 25 (D) 75

40. Medium · Desmos

A rectangular water storage tank at a facility has a base measuring 8 feet by 5 feet. The tank is filled with water to a depth of 4.5 feet. Using a water density of 62.4 pounds per cubic foot, what is the weight of the water in the tank, in pounds?

41. Medium · Desmos

A cylindrical water tank has a base circumference of 44π meters and a height of 15 meters. The volume of the tank is $k\pi$ cubic meters. What is the value of k?

42. Medium · Desmos

A design team is producing rectangular flag replica banners of two sizes. Rectangle A is similar to rectangle B, and the side lengths of rectangle A are 3 times the corresponding side lengths of rectangle B. If the area of rectangle B is 24 square centimeters, what is the area, in square centimeters, of rectangle A?

43. Medium · Desmos

For a school engineering fair project, a team builds a right circular cone-shaped sand mound model with a volume of 48π cubic centimeters and a height of 9 centimeters. What is the radius, in centimeters, of the base of the cone?

44. Medium · Desmos

A cylindrical water tank has a radius of 3 meters and a height of 8 meters. Which of the following is closest to the volume of the tank, in cubic meters? (Use $\pi \approx 3.14$.)

(A) 188.4

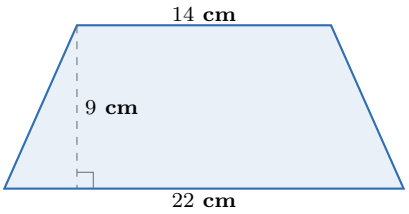
(B) 226.1

(C) 254.3

(D) 150.7

45. Medium · Desmos

A trapezoid has parallel sides of length 14 cm and 22 cm, and a height of 9 cm. What is the area of the trapezoid, in square centimeters?



(A) 144

(B) 153

(C) 162

(D) 171

46. Medium · Desmos

For a particular square, the perimeter is numerically equal to the area. What is the perimeter of the square?

47. Medium · Desmos

A circular coaster has a diameter of 10 cm and a height (thickness) of 3 mm. If 8 coasters are stacked on top of each other, what is the volume of the stack?

(A) 60π cm^3

(B) 75π cm^3

(C) 120π cm^3

(D) 150π cm^3

48. Medium · Desmos

If the surface area of a cube is 96 square centimeters, what is the volume of the cube, in cubic centimeters?

49. Medium · Desmos

Two similar rectangular prisms have volumes 972 and 36. An edge of the larger prism has length 12. What is the length of the corresponding edge of the smaller prism?

50. Medium · Desmos

The edges of cubes A and B are in the ratio $\dfrac{3}{88}$ (edge of A to edge of B). What is the ratio of the surface area of B to the surface area of A?

(A) $\dfrac{9}{7{,}744}$

(B) $\dfrac{3}{88}$

(C) $\dfrac{88}{3}$

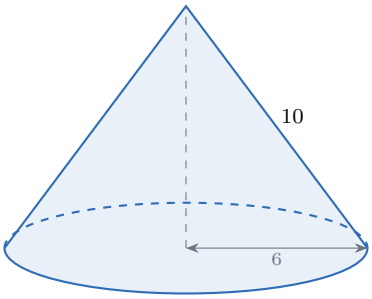
(D) $\dfrac{7{,}744}{9}$

51. Medium · Desmos

A right circular cylinder has volume 90π and height 10. The lateral surface area (excluding the bases) of the cylinder is $k\pi$. What is the value of k?

52. Medium · Desmos

A right circular cone has base radius 6 and slant height 10. The volume of the cone is $k\pi$. What is the value of k?

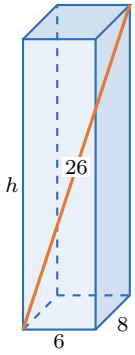


53. Medium · Desmos

A sphere has surface area 144π. The volume of the sphere is $k\pi$. What is the value of k?

54. Medium · Desmos

A rectangular box has a base that measures 6 by 8. The distance from one corner of the box to the opposite corner (the space diagonal) is 26. What is the height of the box?



55. Medium · Desmos

The space diagonal of a cube (the segment joining opposite vertices) has length $9\sqrt{3}$. What is the volume of the cube?

56. Medium · Desmos

A box has interior dimensions 10 cm by 12 cm by 16 cm. Cubes with edge length 2 cm are packed into the box with no gaps. How many cubes fit?

57. Medium · Desmos

A cone and a cylinder each have radius 5 and height 12. The cone is filled with water and the water is poured into the cylinder, repeatedly, until the cylinder is full. How many full cones of water does it take to fill the cylinder?